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Chapter 3

Security Detection Tools

Firewall, WAF, and Proxy

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What is firewall ?

- A **firewall** is a **security device/software** that controls incoming and outgoing network traffic.
- It works like a **security guard** between **trusted network (internal)** and **untrusted network (internet)**.
 - It **allows safe traffic** and **blocks harmful** or unauthorized traffic.
- Firewall: first line of defense.

Purpose of a Firewall

- Protect the network from cyber attacks.
 - Monitor network traffic.
 - Allow approved communication.
 - Block unauthorized access.
 - Prevent malware, exploits, and intrusions.
 - Create security boundaries between networks.

How Does a Firewall Work?

- You give the firewall a set of rules.
- Every incoming/outgoing data packet must pass through the firewall first.
- The firewall
 - ✓ Allows traffic if it matches a safe rule
 - ✗ Blocks traffic if it looks harmful or unauthorized
- Modern firewalls do much more than simple rule checking. They also detect
 - Malware
 - Intrusions
 - Suspicious patterns
 - Dangerous websites

Types of Firewall

- Packet Filtering Firewall.
- Stateful Inspection Firewall.
- Proxy Firewall (Application-Level)
- Next-Generation Firewall (NGFW)
- WAF (Web Application Firewall)

Our focus will be on

- **Proxy Firewall (Application-Level)**
- **WAF (Web Application Firewall)**

1/ Stateless Firewall (Layer 3 & Layer 4)

- Works only using fixed rules (ACLs).
- Does NOT remember past traffic.
- Treats every packet as new, even if it is part of an old connection.

How it works

- Compares each packet with rule list.
- No session tracking.
- Fast but less intelligent.

1/ Stateless Firewall (Layer 3 & Layer 4)

Limitations

- Cannot apply advanced/relationship-based policies.
- Cannot identify if a packet belongs to a good or bad connection.

Example

- If 3 packets from an attacker are blocked, the firewall still checks the 4th packet again instead of automatically blocking it.

Are Stateless Firewall obsolete?

1/ Stateless Firewall (Layer 3 & Layer 4)

Advantages

Stateless firewalls (packet-filtering firewalls) are still used because they are

- **Fast** (no connection tracking)
- **Lightweight** (low CPU/memory usage)
- **Simple** (easy to deploy in routers, switches, embedded systems)

They are still common in

- Core routers (ACLs)
- High-speed networks
- First filtering layer (drop obvious bad traffic)

2/ Stateful Firewall (Layer 3 & Layer 4)

- Works using rules + remembers past connections.
- Maintains a **state table** (connection history).
- More secure than stateless firewalls.

How it works

- If a connection is accepted once, future packets from that connection are auto-allowed.
- If a connection is denied once, future packets from that source are auto-denied.

Advantages

- Smarter decisions
- Recognizes legit sessions
- Blocks malicious repeated attempts

2/ Stateful Firewall (Layer 3 & Layer 4)

- If a connection is accepted, all subsequent packets belonging to that same session are automatically allowed.

Simple analogy

Think of it like this

- Stateless firewall = security guard checking every person every time
- Stateful firewall =
 - Checks you once at entry
 - Gives you a badge
 - Lets you move freely inside **only while the session is valid**

Why stateful firewalls replaced stateless firewalls (in many cases)

Stateful firewalls add

- Connection tracking
- Context awareness
- Better security decisions

So today

- **Stateful = standard baseline**
- **Stateless = supporting role**

Modern networks can use **both**

1. **Stateless firewall (ACL)**

→ Drop clearly unwanted traffic early (cheap & fast)

2. **Stateful firewall**

→ Perform deeper inspection and session validation

3/ Proxy Firewall (Application-Level Gateway) Layer 7

- Works as a **middleman** between user and the internet.
 - Acts as an intermediary: the client connects to the proxy, and the proxy connects to the server.
- Can inspect the actual content inside packets.

What it can do

- Understand HTTP, FTP, DNS, etc.
- Provide content filtering (block websites, keywords).
- Hide internal IP addresses → gives anonymity (provides privacy, not full anonymity)
- Can inspect application-layer content (e.g., HTTP requests, URLs, payloads)

Best for

- Web filtering (WHAT you can access)
- Web browsing control (HOW and WHEN you can access it)
- Extra secure environments (Banks and financial institutions; Government or military networks; Healthcare systems (sensitive data); Corporate networks with strict policies; Research labs with confidential data)

3/ Proxy Firewall (Application-Level Gateway) Layer 7

Web filtering (content-level)

This is about **what content is allowed or blocked**.

Examples

- Block specific websites (e.g., social media)
- Block categories (gambling, malware sites, etc.)
- Filter keywords in URLs or pages
- Block malicious or phishing pages

Focus: **content & websites**

3/ Proxy Firewall (Application-Level Gateway) Layer 7

Web browsing control (behavior-level)

This is about **how users browse** the web.

Examples:

- Allow browsing only during certain hours (e.g., 8am–6pm)
- Block file downloads (e.g., **.exe**, **.zip**)
- Limit access based on user roles (students vs admins)
- Force authentication before accessing the internet

Focus: **user behavior & policies**

4/ Next-Generation Firewall (NGFW) – Layer 3 to Layer 7

Most Advanced Firewall

NGFW combines all previous firewall features + many security protections.

A **NGFW** combines

- Stateless + Stateful + Proxy features
- PLUS advanced security mechanisms

It doesn't just allow/block traffic. It **understands, analyzes, and defends**

4/ Next-Generation Firewall (NGFW) – Layer 3 to Layer 7

Key Features

- Deep Packet Inspection (DPI)
 - Looks **inside packets**, not just headers. Can detect hidden malware or suspicious content
- Intrusion Prevention System (IPS)
 - Detects and automatically blocks attacks in real time E.g., stops exploitation attempts or known attack signatures
- Heuristic Analysis (detects attack patterns)
 - Detects unknown or new attacks based on behavior/patterns. E.g.: unusual traffic = possible attack
- SSL/TLS Decryption (inspects encrypted traffic)
 - Decrypts HTTPS traffic to inspect it. Important because most attacks are now hidden in encrypted traffic
- Application Control (identify Facebook, YouTube, SSH, etc.)
 - Identifies and controls specific applications, not just ports. E.g.: allow YouTube but block Facebook
- Threat Intelligence Integration
 - Uses external threat databases (updated constantly). Automatically blocks known malicious IPs, domains, signatures

4/ Next-Generation Firewall (NGFW) – Layer 3 to Layer 7

- Why it is the best
 - Blocks attacks in real-time
 - Detects anomalies
 - Understands applications
 - Most suitable for modern networks

Scenario: Malicious File Download via HTTPS

A user tries to download a file from a website

<https://example-download.com/update.exe>

The file actually contains **malware**.

Case 1: Stateful Firewall (Layer 3/4)

What it sees

- Source IP
- Destination IP
- Port 443 (HTTPS)
- Connection is established

Decision

- “This is a valid HTTPS connection” → **ALLOW**

Problem

- It **cannot see inside HTTPS**
- Malware passes through undetected

Case 2: Proxy Firewall (Layer 7)

What it sees

- Can inspect HTTP/HTTPS (if configured)

Possible actions

- Block based on URL or category
- Scan file (limited)
 - a. Check file type (block .exe, allow .pdf)
 - b. Compare file against a basic signature database
 - c. Apply simple rules (size, extension, source URL)

Limitation

- May **not detect sophisticated malware**
- SSL inspection may not always be enabled

Case 3: NGFW (Next-Generation Firewall)

What it does

1. **SSL/TLS Decryption**
 - Opens HTTPS traffic
2. **Deep Packet Inspection (DPI)**
 - Inspects file content
3. **Heuristic Analysis**
 - Detects suspicious behavior in the file
4. **Intrusion Prevention System (IPS)**
 - Matches known attack signatures
5. **Threat Intelligence**
 - Checks if the website is malicious

Final Decision: BLOCK the download

Scenario: Malicious File Download via HTTPS

Stateful firewall → “Connection looks OK”

Proxy firewall → “Website looks suspicious”

NGFW → “File is malware”

- A traditional firewall protects **connections**
- A NGFW protects **content, applications, and users**
- “HTTPS made old firewalls blind. NGFW gives them eyes again.”

Comparison Table

Stateless Firewall	- Basic filtering - No memory of past connections - Very fast, good for high-speed networks
Stateful Firewall	- Understands traffic patterns - Maintains connection history - Can apply complex rules
Proxy Firewall	- Inspects packet contents - Supports content filtering - Application control - Masks internal IPs
NGFW (Next-Generation)	- Advanced threat protection - Built-in IPS - Heuristic/anomaly detection - Decrypts & inspects SSL/TLS traffic

End of Lecture Questions?

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Proxy

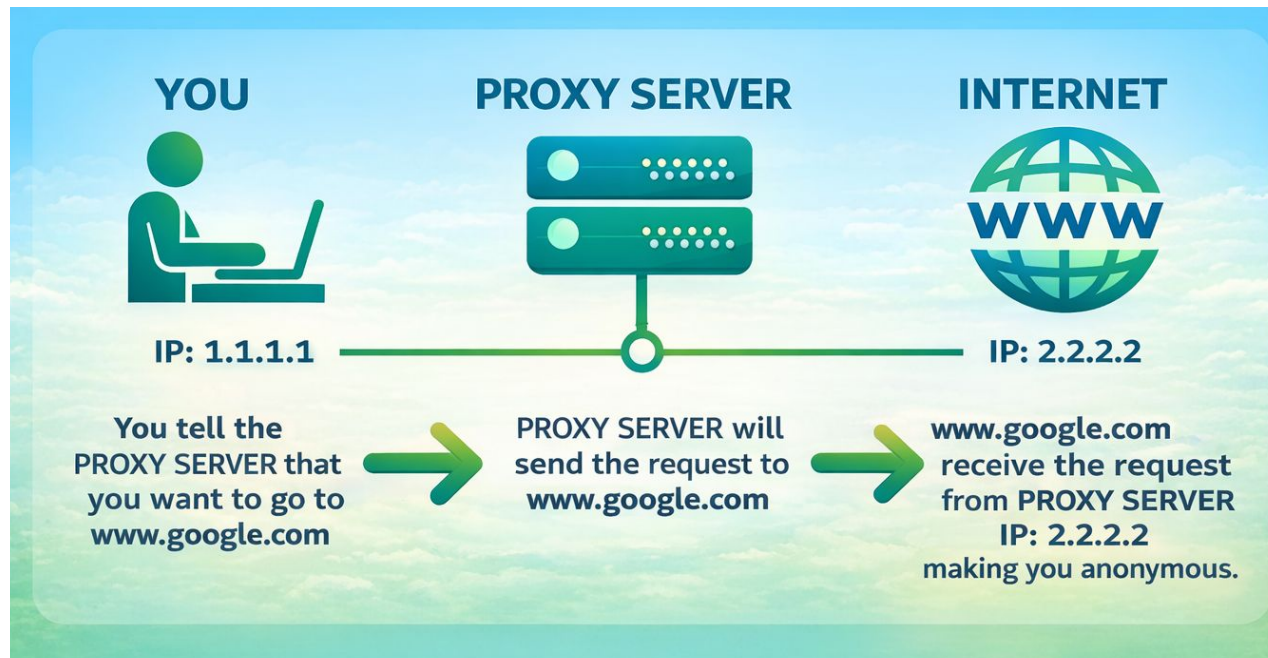
Outline

1. Definition of a Proxy Server
2. How a Proxy Server Works
3. Types of Proxy Servers
4. Log Formats Supported by Proxies
5. Logs Generated and Sent to SIEM
6. Advantages of Using a Proxy Server for Monitoring
7. Disadvantages of Proxy Servers
8. Proxy Log Flow to SIEM
9. Top Proxy Vendors in 2026
10. Top use-cases for Proxy Server Monitoring

1. Definition of a Proxy Server

A **Proxy Server** is an **intermediary system** that sits between **user devices (e.g., employees)** and the **internet** to monitor, filter, log, and control web traffic.

It is used to monitor employee activity, block malicious sites, enforce security policies, and ensure compliance with company regulations.



2. How a Proxy Server Works

Step-by-Step Process

1. **User Request** → An employee tries to access a website.
2. **Proxy Interception** → The proxy captures the request.
3. **Policy Enforcement** → The proxy checks company security rules.
 - o **Allowed?** → Request is forwarded to the internet.
 - o **Blocked?** → The request is denied, and the user gets a block page.
4. **Traffic Logging** → The proxy logs the user's activity.
5. **Response Handling** → If allowed, the website's response is sent back to the employee.

3. Types of Proxy Servers

- 1) Forward Proxy
- 2) Reverse Proxy
- 3) Transparent Proxy

3. Types of Proxy Servers

1) Forward Proxy

- **Usage:** Controls and monitors employee outbound internet traffic.
- **Tools:** Squid Proxy, Blue Coat (Symantec ProxySG).
- **SOC Relevance:** Helps in web traffic filtering, blocking malicious domains, and generating logs forwarded to SIEM for monitoring insider threats or policy violations.

2) Reverse Proxy

- **Usage:** Protects and load-balances internal web servers (e.g., APIs, websites).
- **Tools:** NGINX, Apache HTTPD, F5 BIG-IP, Citrix ADC.
- **SOC Relevance:** Key for hiding backend infrastructure, terminating SSL, and mitigating DDoS. Logs help track external client access to internal resources.

3. Types of Proxy Servers

3) Transparent Proxy

- **Usage:** Enforces policies without user configuration.
- **Tools:** Cisco Web Security Appliance (WSA), FortiGate, Palo Alto firewalls (URL filtering).
- **SOC Relevance:** Passive monitoring of web traffic for threat intel, malware callouts, or data exfiltration.

4. Log Formats Supported by Proxies

Proxy servers generate logs in multiple formats to integrate with SIEM, DLP, and network monitoring tools:

- Syslog (RFC 5424 format)
- Common Event Format (CEF)
- Log Event Extended Format (LEEF)
- JSON (JavaScript Object Notation)
- Plain Text (CSV, TXT)

5. Logs Generated and Sent to SIEM

Common Proxy Logs Sent to SIEM

- **Website Access Logs** – Records URLs visited by employees.
- **Blocked Request Logs** – Shows attempts to access restricted websites.
- **Bandwidth Usage Logs** – Logs large file downloads or excessive streaming.
- **User Authentication Logs** – Tracks who accessed which sites.
- **Malware Detection Logs** – Logs access to suspicious or phishing domains.

6. Advantages of Using a Proxy Server for Monitoring

1. **Enhances Security** – Blocks malicious sites and phishing attempts.
2. **Reduces Data Leakage** – Prevents unauthorized uploads of sensitive data.
3. **Controls Employee Internet Usage** – Limits access to non-work-related websites.
4. **Improves Bandwidth Management** – Restricts excessive streaming or downloads.
5. **Enforces Compliance** – Helps meet PCI DSS, GDPR, HIPAA security policies.
6. **Speeds Up Content Delivery** – Uses caching to load frequently visited sites faster.
7. **Protects Against Insider Threats** – Detects unusual web activity.

7. Disadvantages of Proxy Servers

1. **May Slow Down Internet Speed** – Extra processing time for filtering traffic.
2. **Bypass Risks** – Users may use VPNs or proxy avoidance tools.
Examples?
3. **Privacy Concerns** – Employees might feel monitored too much.
4. **Maintenance Overhead** – Requires updates and tuning to keep effective.

8. Proxy Log Flow to SIEM

1. **Employee Sends a Request** → Tries to access a website.
2. **Proxy Captures and Logs the Request** → Checks against security rules.
3. **Logs Forwarded to SIEM** → SIEM correlates with other security events.
4. **SOC Team Investigates Anomalies** → If suspicious behavior is detected, action is taken.

Example SIEM Use Case

- **Scenario:** Employee **uploads sensitive data** to an unknown website.
- Proxy Server Action:
 - o Detects and **blocks the request**.
 - o Sends an alert to **SIEM**.
 - o SOC analysts investigate **data exfiltration risk**.

9. Top Proxy Vendors (2025-26)

1. Zscaler Internet Access (ZIA)
2. Blue Coat (Broadcom Symantec)
3. Palo Alto Prisma Access
4. Cisco Umbrella
5. Forcepoint Web Security
6. McAfee Web Gateway
7. Squid Proxy (Open-Source)
8. Fortinet FortiProxy
9. Sophos Web Gateway
10. Barracuda Web Security Gateway

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Web Application Firewall (WAF)

Layer 7 Security for Web Applications

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What is WAF ?

A web application firewall (WAF) is an application firewall for HTTP applications.

A WAF operates through a set of rules often called policies. These policies aim to protect against vulnerabilities in the application by filtering out malicious traffic.

By deploying a WAF in front of a web application, a shield is placed between the web application and the Internet.

While a proxy server protects a client machine's identity by using an intermediary, a WAF is a type of reverse-proxy, protecting the server from exposure by having clients pass through the WAF before reaching the server.

What is WAF ?



WAF protects web applications from a variety of application layer attacks such as cross-site scripting (XSS), SQL injection, and cookie poisoning, among others.

Attacks to apps are the leading cause of breaches—they are the gateway to your valuable data.

With the right WAF in place, you can block the array of attacks that aim to exfiltrate that data by compromising your systems.

Next Lecture will be dedicated to studying WAFs in depth.